

Exercice I.

[20 points]

Calculer les primitives suivantes:

1. $\int \cot x \, dx,$

2. $\int \cos(\sqrt{x}) \, dx,$

3. $\int \frac{1}{2 + \sin x} \, dx,$

4. $\int \frac{3x + 1}{\sqrt{x^2 + x + 1}} \, dx.$

Exercice II.

[10 points]

Sans calculer les coefficients, donner la forme générale de la décomposition en facteurs simples pour la fonction rationnelle suivante:

$$f(x) = \frac{x^2 + 1}{x(x-1)^2(x^2 + 2x + 3)^2}.$$

Exercice III.

[25 points]

1. Calculer les limites suivantes:

a. $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k^2}{n^3} \ln \left(1 + \frac{k}{n} \right)$

b. $\lim_{x \rightarrow 0^+} \int_x^{2x} \frac{e^{t+1}}{t} \, dt$

2. Soit $g(x) = \int_{x^2-1}^{x^2} \ln t \, dt$. Calculer $g'(x)$ et spécifier le domaine de dérivabilité de g .

Exercice IV.

[25 points]

1. Etudier la nature des intégrales suivantes: $I = \int_0^{+\infty} \frac{x \ln x}{1+x^3} \, dx$ et $J = \int_1^3 \frac{1}{\sqrt{(x-1)(3-x)}} \, dx$.

2. (a) Montrer que l'intégrale impropre suivante $K = \int_1^{+\infty} \cos(2e^x) \, dx$ est convergente.

(b) Dédire que l'intégrale suivante $L = \int_1^{+\infty} \frac{1}{\sqrt{x}} \cos(2e^x) \, dx$ est convergente.

(c) Dédire la nature de $M = \int_1^{+\infty} \left| \frac{1}{\sqrt{x}} \cos(e^x) \right| \, dx$.

Exercice V.

[20 points]

Soit $I_n = \int_0^{\infty} x^{2n} e^{-\frac{x^2}{2}} \, dx$. On admet que $I_0 = \int_0^{\infty} e^{-\frac{x^2}{2}} \, dx = \sqrt{\frac{\pi}{2}}$.

1. Montrer que l'intégrale impropre I_n est convergente $\forall n \in \mathbb{N}$.

2. Calculer $\int x e^{-\frac{x^2}{2}} \, dx$.

3. Montrer que $I_n = (2n-1)I_{n-1}$, $\forall n \in \mathbb{N}^*$.

4. Dédire la valeur de I_n .



20 Exercise I. Calculate the following integrals.

$$1. \int \cot x \, dx = \int \frac{\cos x}{\sin x} \, dx \stackrel{(2.5)}{=} \int \frac{d(\sin x)}{\sin x} \stackrel{(2.5)}{=} \ln|\sin x| + C; \quad C \in \mathbb{R}.$$

$$2. \int \cos(\sqrt{x}) \, dx \stackrel{(1)}{=} 2 \int t \cos t \, dt \stackrel{(3)}{=} 2t \sin t - 2 \cos t + C; \quad C \in \mathbb{R}$$

$$\sqrt{x} = t \Rightarrow x = t^2 \quad \left| \begin{array}{l} t \xrightarrow{\cos t} \cos t \\ 1 \xrightarrow{\sin t} \sin t \\ 0 \xrightarrow{-\cos t} -\cos t \end{array} \right. \\ \Rightarrow dx = 2t \, dt$$

$$3. \int \frac{1}{2 + \sin x} \, dx \stackrel{(1)}{=} \int \frac{1}{t^2 + t + 1} \, dt$$

$$t = \tan \frac{x}{2} \Rightarrow dx = \frac{2 \, dt}{1+t^2} \quad \text{of} \quad \sin x = \frac{2t}{1+t^2}$$

$$\stackrel{(1)}{=} \int \frac{1}{(t + \frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} \, dt = \frac{2}{\sqrt{3}} \arctan \frac{2t+1}{\sqrt{3}} + C \stackrel{(1)}{=} \frac{2}{\sqrt{3}} \arctan \left(\frac{2 \tan \frac{x}{2} + 1}{\sqrt{3}} \right) + C.$$

$$4. \int \frac{3x+1}{\sqrt{x^2+x+1}} \, dx \stackrel{(1)}{=} \frac{3}{2} \int \frac{2(x + \frac{1}{2})}{\sqrt{x^2+x+1}} \, dx \stackrel{(1)}{=} \frac{3}{2} \int \frac{2x+1}{\sqrt{x^2+x+1}} \, dx - \frac{1}{2} \int \frac{1}{\sqrt{x^2+x+1}} \, dx \\ \stackrel{(1)}{=} \frac{3}{2} \int \frac{d(x^2+x+1)}{\sqrt{x^2+x+1}} - \frac{1}{2} \int \frac{d(x + \frac{1}{2})}{\sqrt{(x + \frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2}} \\ \stackrel{(1)}{=} \frac{3}{2} \cdot 2 \sqrt{x^2+x+1} - \frac{1}{2} \ln \left((x + \frac{1}{2}) + \sqrt{x^2+x+1} \right) + C$$

10 Exercise II. Give the general form of the partial fraction decomposition:

$$f(x) = \frac{x^2+1}{x(x-1)^2(x^2+2x+3)^2}$$

$$= \frac{a}{x} + \frac{b}{(x-1)} + \frac{c}{(x-1)^2} + \frac{dx+e}{(x^2+2x+3)} + \frac{gx+h}{(x^2+2x+3)^2}. \quad (2 \times 5)$$

25 Exercise III.

1. Calculate the following limits:

$$a. \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{k^2}{n^2} \ln \left(1 + \frac{k}{n} \right) \stackrel{(2)}{=} \int_0^1 \frac{x^2 \ln(1+x)}{u} \, dx \quad (\because f(x) = x^2 \ln(1+x) \stackrel{(1)}{\text{is cont on } [0,1]}, \text{ hence integrable on } [0,1])$$

$$\stackrel{(1)}{=} \frac{x^3}{3} \ln(1+x) \Big|_0^1 - \int_0^1 \frac{x^3}{3} \cdot \frac{1}{1+x} \, dx$$

$$= \frac{\ln 2}{3} - \frac{1}{3} \int_0^1 \frac{x^3}{1+x} \, dx$$

$$= \frac{\ln 2}{3} - \frac{1}{3} \int_0^1 \left[x^2 - x + 1 + \frac{1}{x+1} \right] \, dx$$

$$\stackrel{(1)}{=} \frac{\ln 2}{3} - \frac{1}{3} \left[\frac{x^3}{3} - \frac{x^2}{2} - x + \ln(x+1) \right]_0^1$$

$$= \frac{\ln 2}{3} - \frac{1}{3} \left[\left(\frac{1}{3} - \frac{1}{2} - 1 + \ln 2 \right) - (0) \right] \\ \stackrel{(1)}{=} \frac{2}{3} \ln 2.$$

$$\begin{array}{r} x^2 + x + 1 \\ x + 1 \overline{) x^3 + x^2 + x + 1} \\ \underline{x^3 + x^2} \\ x \\ \underline{x + 1} \\ 0 \end{array}$$

$$b. \lim_{x \rightarrow 0^+} \int_x^{2x} \frac{e^{t+1}}{t} dt$$

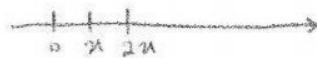
$$\stackrel{M.V.T}{=} \lim_{x \rightarrow 0^+} e^{c+1} \int_x^{2x} \frac{1}{t} dt ;$$

$$\stackrel{(1)}{=} e \ln 2.$$

$$\stackrel{(1)}{x} \leq c \leq 2x \\ \text{as } x \rightarrow 0^+, c \rightarrow 0^+$$

$$\textcircled{1} f(t) = e^{t+1} \text{ is cont on } [x, 2x]$$

$$\textcircled{1} g(t) = \frac{1}{t} > 0 \text{ on } [x, 2x]$$



2. Let $g(x) = \int_{x^2-1}^{x^2} \ln t dt$. Compute g' & specify the domain of diff of g .

$$g'(x) = (x^2)' \ln x^2 - (x^2-1)' \ln (x^2-1) = 2x \ln x^2 - 2x \ln (x^2-1).$$

• Domain of diff of g , $I = ?$

$$\textcircled{1} f(t) = \ln t \text{ is cont on }]0, \infty[= D$$

$$\textcircled{1} u(x) = x^2 \text{ \& } v(x) = x^2 - 1 \text{ diff on } \mathbb{R} = I_1$$

$$u(x) = x^2 \in D =]0, \infty[\Leftrightarrow x \in]-\infty, 0[\text{ or } x \in]0, +\infty[.$$

$$v(x) = x^2 - 1 \in D =]0, +\infty[\Leftrightarrow (x-1)(x+1) > 0 \Leftrightarrow x \in]-\infty, -1[\text{ or } x \in]1, +\infty[$$

$$\textcircled{1} u, v \in D \Leftrightarrow x \in I_2, I_2 =]-\infty, -1[\cup]1, +\infty[.$$

$$\text{Now } \textcircled{2} I = I_1 \cap I_2 =]-\infty, -1[\cup]1, +\infty[.$$

Exercise IV. Study the nature of the following imp integrals:

$$1. I = \int_0^{+\infty} \frac{x \ln x}{1+x^3} dx ; f \in C([0, \infty)) \Rightarrow f \in R_{loc}([0, \infty))$$

$$0 < \frac{x \ln x}{1+x^3} \underset{\text{near } 0}{\sim} \frac{x \ln x}{x^3} \underset{\infty}{\sim} \frac{x \ln x}{x^3} = \frac{1}{x^2 \ln' x}. \text{ I is conv (using Bertrand criteria } x=2 \text{ \& equivalent test)}$$

$$\text{OR } x^{3/2} \cdot \frac{x \ln x}{1+x^3} \underset{\infty}{\sim} \frac{\ln x}{x^{1/2}} \xrightarrow{x \rightarrow \infty} 0 ; \alpha = 3/2 > 1 \text{ imply I is abs conv.}$$

$$J = \int_1^3 \frac{1}{\sqrt{(x-1)(3-x)}} dx ; f \in C(]1, 3[) \Rightarrow f \in R_{loc}(]1, 3[).$$

$$= \int_1^2 \frac{1}{\sqrt{(x-1)(3-x)}} dx + \int_2^3 \frac{1}{\sqrt{(x-1)(3-x)}} dx \stackrel{(1)}{=} J_1 + J_2$$

$$\text{• For } J_1, \text{ near } 1^+ : f(x) = \frac{1}{\sqrt{(x-1)(3-x)}} \stackrel{(1)}{\sim} \frac{1}{\sqrt{(x-1)2}} = \frac{1}{\sqrt{2}} \frac{1}{(x-1)^{1/2}}$$

$$J_1 \text{ conv (Riemann ex } \alpha = 1/2 < 1)$$

• For J_2 , near 3^- : $\mathcal{O}(n) = \frac{1}{\sqrt{(n-1)(3-n)}} \stackrel{\textcircled{1}}{\sim} \frac{1}{\sqrt{2(3-n)}} = \frac{1}{\sqrt{2}} \frac{1}{(3-n)^{1/2}}$
 J_2 conv (Riemann $\alpha = 1/2 < 1$).

Thus j is convergent.

2. (a) $K = \int_1^\infty \cos(2e^x) dx$; $f \in C([1, \infty)) \Rightarrow f \in R_{loc}([1, \infty))$.

$K \stackrel{\textcircled{1}}{=} \int_1^\infty \underbrace{\frac{1}{2e^x}}_g \underbrace{\cos(2e^x)(2e^x)}_f dx$

• $g \downarrow$ & $g(n) \xrightarrow{n \rightarrow \infty} 0$

• $|F(n)| = \left| \int_1^n \cos(2e^x) d(2e^x) \right|$
 $= |\sin(2e^n) - \sin(2e)|$
 $\leq |\sin(2e^n)| + |\sin(2e)|$
 $\stackrel{\textcircled{1}}{\leq} 1 + 1 = 2 \text{ on } [1, \infty)$

$\Rightarrow K$ is convergent (Abel-Dirichlet)

or $K \stackrel{\textcircled{1}}{=} \int_{2e}^\infty \underbrace{\frac{1}{t}}_g \underbrace{\cos t}_f dt$

; $2e^x = t \Rightarrow 2e^x dx = dt$
 $\Rightarrow dx = \frac{1}{t} dt$

• $g \downarrow$ & $g(t) \xrightarrow{t \rightarrow \infty} 0$

• $|F(n)| = \left| \int_{2e}^n \cos t dt \right| \leq 2$

$\Rightarrow K$ is conv (Abel-Dirichlet)

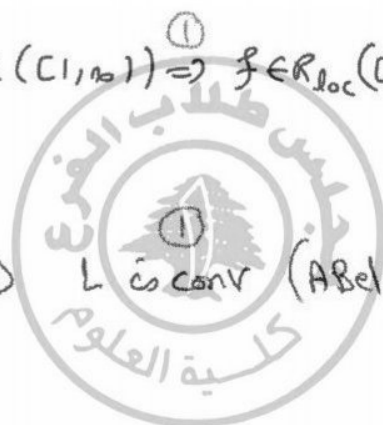
or using integration by parts $K = \int_1^\infty \underbrace{\frac{1}{2e^x}}_v \underbrace{\cos(2e^x)(2e^x)}_{u'} dx$

(b). $L = \int_1^{+\infty} \frac{1}{\sqrt{x}} \cos(2e^x) dx$; $f \in C([1, \infty)) \stackrel{\textcircled{1}}{\Rightarrow} f \in R_{loc}([1, \infty))$

• $g(x) = \frac{1}{\sqrt{x}} \downarrow$ & $\frac{1}{\sqrt{x}} \xrightarrow{x \rightarrow \infty} 0$

• $F(x) \stackrel{\textcircled{1}}{=} \int_1^x \cos(2e^t) dt$ is bdd
 (since $\mathcal{O}$ is conv)

$\Rightarrow L$ is conv (Abel-Dirich)



or $\bullet g(n) = \frac{1}{\sqrt{x}} \downarrow \& b.d. \text{ on } [1, +\infty[$
 $\bullet \int_1^{+\infty} \cos(2e^x) dx \text{ conv} \quad \} \Rightarrow L \text{ is conv (Abel-Dini)}$

(c) $M = \int_1^{+\infty} \left| \frac{1}{\sqrt{x}} \cos(e^x) \right| dx$

$\frac{|\cos e^x|}{\sqrt{x}} \stackrel{(1)}{\geq} \frac{\cos^2(e^x)}{x^{1/2}} \stackrel{(1)}{\geq} \frac{1 + \cos(2e^x)}{2 x^{1/2}} = \frac{1}{2} \left[\frac{1}{x^{1/2}} + \frac{\cos(2e^x)}{x^{1/2}} \right]$

$\bullet \int_1^{\infty} \frac{1}{x^{1/2}} dx \text{ div (Riemann } \alpha = 1/2 < 1)$

$\bullet \int_1^{\infty} \frac{1}{\sqrt{x}} \cos(2e^x) dx \text{ conv (using } b_0)$

Hence M is divergent (using comparison test).

20 Exercise V. let $I_n = \int_0^{\infty} x^{2n} e^{-\frac{x^2}{2}} dx$. Given $I_0 = \int_0^{\infty} e^{-\frac{x^2}{2}} dx = \sqrt{\pi/2}$

1. $x^{10} \cdot \frac{x^{2n}}{e^{x^2/2}} \xrightarrow{x \rightarrow \infty} 0$ & $\alpha = 10 > 1 \Rightarrow I_n \text{ conv } \forall n \in \mathbb{N}$.

2. $\int x e^{-\frac{x^2}{2}} dx = -\int e^{-\frac{x^2}{2}} d\left(-\frac{x^2}{2}\right) = -e^{-\frac{x^2}{2}} + C; C \in \mathbb{R}$.

3. $I_n = \int_0^{\infty} \frac{x^{(2n-1)}}{u} \frac{x e^{-\frac{x^2}{2}}}{v'} dx \quad \text{By parts}$
 $= x^{2n-1} \left(-e^{-\frac{x^2}{2}} \right) \Big|_0^{\infty} - \int_0^{\infty} (2n-1) x^{2n-2} \left(-e^{-\frac{x^2}{2}} \right) dx$
 $= (2n-1) \int_0^{\infty} x^{2(n-1)} e^{-\frac{x^2}{2}} dx$
 $= (2n-1) I_{n-1} \quad \forall n \in \mathbb{N}^*$

or we integrate I_{n-1} by parts to obtain $I_n = (2n-1) I_{n-1}$.

4. $I_n = (2n-1) I_{n-1} \quad (1)$
 $= (2n-1)(2n-3) I_{n-2} \quad (1)$
 $= (2n-1)(2n-3)(2n-5) \dots 3 \cdot 1 \cdot I_0$
 $= 1 \cdot 3 \cdot 5 \dots (2n-3)(2n-1) \cdot \sqrt{\pi/2} \quad (3)$

